

Chapter 5: Programming

5.1 Create / Select Programs for Editing

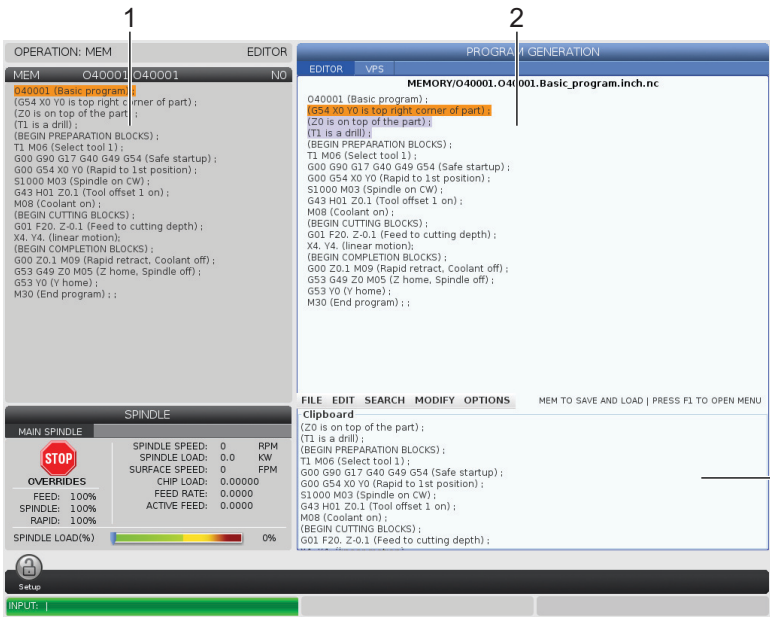
You use the Device Manager ([LIST PROGRAM]) to create and select programs for editing. Refer to page 85 to create a new program. Refer to page 86 to choose an existing program to edit.

5.2 Program Edit Modes

The Haas control has (2) program edit modes: The program editor or manual data input (MDI). You use the program editor to make changes to numbered programs stored in an attached memory device (machine memory, USB, or net share). You use MDI mode to command the machine without a formal program.

The Haas control screen has (2) program edit panes: The Active Program / MDI pane, and the Program Generation pane. The Active Program / MDI pane is on the left side of the screen in all display modes. The Program Generation pane appears only in EDIT mode.

F5.1: Example Edit Panes. [1] Active Program / MDI Pane, [2] Program Edit Pane, [3] Clipboard Pane



5.2.1 Basic Program Editing

This section describes the basic program editing functions. These functions are available when you edit a program.

1. To write a program, or make changes to a program:
 - a. To edit a program in MDI, press **[MDI]**. This is **EDIT:MDI** mode. The program is displayed on the Active pane.
 - b. To edit a numbered program, select it in the Device Manager (**[LIST PROGRAM]**), then press **[EDIT]**. This is **EDIT:EDIT** mode. The program is displayed in the Program Generation pane.
2. To highlight code:
 - a. Use the cursor arrow keys or the jog handle to move the highlight cursor through the program.
 - b. You can interact with single pieces of code or text (cursor highlighting), blocks of code, or multiple blocks of code (block selection). Refer to the Block Selection section for more information.
3. To add code to the program:
 - a. Highlight the code block you want the new code to follow.
 - b. Type the new code.
 - c. Press **[INSERT]**. Your new code appears after the block you highlighted.
4. To replace code:
 - a. Highlight the code you want to replace.
 - b. Type the code you want to replace the highlighted code with.
 - c. Press **[ALTER]**. Your new code takes the place of the code you highlighted.
5. To remove characters or commands:
 - a. Highlight the text you want to delete.
 - b. Press **[DELETE]**. The text that you highlighted is removed from the program.
6. Press **[UNDO]** to reverse up to the last (40) changes.



NOTE:

*You cannot use **[UNDO]** to reverse changes that you made if you exit **EDIT:EDIT** mode.*



NOTE:

*In **EDIT:EDIT** mode, the control does not save the program as you edit. Press **[MEMORY]** to save the program and load it into the Active Program pane.*

Block Selection

When you edit a program, you can select single or multiple blocks of code. You can then copy and paste, delete, or move those blocks in one step.

To select a block:

1. Use the cursor arrow keys to move the highlight cursor to the first or last block in your selection.



NOTE:

You can start a selection at the top block or the bottom block, and then move up or down as appropriate to complete your selection.



NOTE:

You cannot include the program name block in your selection. The control gives the message `GUARDED CODE`.

2. Press **[F2]** to start your selection.
3. Use the cursor arrow keys or the jog handle to expand the selection.
4. Press **[F2]** to complete the selection.

Actions with a Block Selection

After you make a text selection, you can copy and paste it, move it, or delete it.



NOTE:

These instructions assume you have already made a block selection as described in the Block Selection section.



NOTE:

*These are actions available in MDI and the Program Editor. You cannot use **[UNDO]** to reverse these actions.*

1. To copy and paste the selection:
 - a. Move the cursor to the location where you want to put a copy of the text.
 - b. Press **[ENTER]**.

The control puts a copy of the selection on the next line after the cursor location.

**NOTE:**

The control does not copy the text to the clipboard when you use this function.

2. To move the selection:
 - a. Move the cursor to the location where you want to move the text.
 - b. Press **[ALTER]**.

The control removes the text from its current location and puts it in the line after the current line.

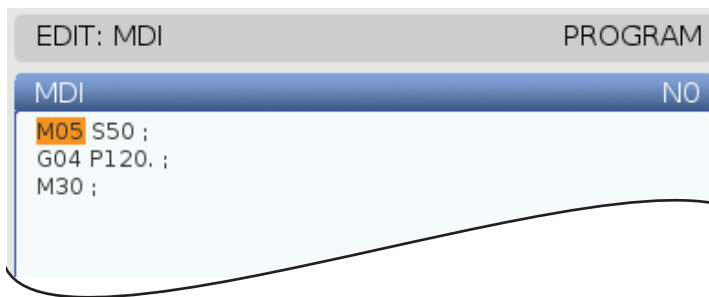
3. Press **[DELETE]** to delete the selection.

5.2.2 Manual Data Input (MDI)

Manual Data Input (MDI) lets you command automatic CNC moves without a formal program. Your input stays on the MDI input page until you delete it.

F5.2:

MDI Input Page Example



1. Press **[MDI]** to enter **MDI** mode.
2. Type your program commands in the window. Press **[CYCLE START]** to execute the commands.
3. If you want to save the program you created in MDI as a numbered program:
 - a. Press **[HOME]** to put the cursor at the beginning of the program.
 - b. Type a new program number. Program numbers must follow the standard program number format (Onnnnnn).
 - c. Press **[ALTER]**.
 - d. In the RENAME pop-up window, you can type a filename and file title for the program. Only the O number is required.
 - e. Press **[ENTER]** to save the program to memory.
4. Press **[ERASE PROGRAM]** to delete everything from the MDI input page.

5.2.3 Program Editor

The Program Editor is a full-featured editing environment with access to powerful functions in an easy-to-use pull-down menu. You use the Program Editor for normal editing.

Press **[EDIT]** to enter edit mode and use the Program Editor.

F5.3: Example Program Editor Screen. [1] Main Program Display, [2] Menu Bar, [3] Clipboard



Program Editor Pull-Down Menu

The Program Editor uses a pull-down menu to give you easy access to editor functions in (5) categories: **File**, **Edit**, **Search**, and **Modify**. This section describes the categories and the choices you can make when you select them.

To use the pull-down menu:

1. Press **[EDIT]** to start the Program Editor.
2. Press **[F1]** to access the pull-down menu.

The menu opens for the last category you used. If you have not yet used the pull-down menu, the **File** menu opens by default.
3. Use the **[LEFT]** and **[RIGHT]** cursor arrow keys to highlight a category. When you highlight a category, the menu appears below the category name.

4. Use the **[UP]** and **[DOWN]** cursor arrow keys to select an option inside of the current category.
5. Press **[ENTER]** to execute the command.

Some menu commands require further input or confirmation. In these cases, an input window or confirmation pop-up appears on the screen. Type your input in the field(s) where applicable, and then press **[ENTER]** to confirm the action or **[UNDO]** to close the pop-up and cancel the action.

File Menu

The **F**ile menu has these options:

- **New**: Creates a new program. In the pop-up menu fields, type an O number (required), a filename (optional), and a file title (optional). For more information about this menu, refer to “Create a New Program” in the Operation section of this manual.
- **Set To Run**: Saves the program and puts it in the active program pane on the left-hand side of the screen. You can also press **[MEMORY]** to use this function.
- **Save**: Saves the program. The program’s filename and path change from red to black to show that the changes are saved.
- **Save As**: You can save the file under any filename. The program’s new filename and path will change from red to black to show that the changes are saved.
- **Discard Changes**: Reverts any changes you’ve made since the file was last saved.

Edit Menu

The **E**dit menu has these options:

- **Undo**: Reverses the last edit operation, up to the last (40) edit operations. You can also press **[UNDO]** to use this function.
- **Redo**: Reverses the last undo operation, up to the last (40) undo operations.
- **Cut Selection To Clipboard**: Removes the selected lines of code from the program and puts them in the clipboard. Refer to “Block Selection” to learn how to make a selection.
- **Copy Selection To Clipboard**: Puts the selected lines of code in the clipboard. This operation does not remove the original selection from the program.
- **Paste From Clipboard**: Puts a copy of the clipboard contents below the current line. This does not remove the clipboard contents.
- **Insert File Path (M98)**: Lets you select a file from a directory and creates the path with the M98.
- **Insert Media File (M130)**: Lets you select a media file from a directory and creates the path with the M130.
- **Special Symbols**: Inserts a special symbol.

Search Menu

The **Search** menu gives you access to the **Find And Replace Text** function. This function lets you quickly find code in the program and optionally replace it. To use it:



NOTE:

This function searches for program code, not text. You cannot use this function to find text strings (like comments).

F5.4:

Find and Replace Menu Example: [1] Text to Find, [2] Replacement Text, [3] Search Direction, [4] Find Option, [5] Replace Option, [6] Find and Replace Option, [7] Replace all Option

The screenshot shows a menu titled "FIND AND REPLACE TEXT" with a blue header. Below the header are several input fields and buttons. Numbered callouts point to specific elements: 1 points to the "Find:" text box, 2 points to the "Replace:" text box, 3 points to the "Direction:" section which includes "Forward" (selected with a radio button) and "Backward" (unselected). 4 points to the "Find (F1)" button, 5 points to the "Replace (F2)" button, 6 points to the "Find/Replace (F3)" button, and 7 points to the "Replace All (Can't UNDO!) (F4)" button.

Specify Your Find / Replace Code

1. Press **[ENTER]** in the editor pull-down menu to open the **Find And Replace Text** menu. Use the cursor arrow keys to move between the fields in the menu.
2. In the **Find** field, Type the code you want to search for.
3. If you want to replace some or all of the found code, type the replacement code in the **Replace** field.
4. Use the **[LEFT]** and **[RIGHT]** cursor arrow keys to choose the search direction. **Forward** searches the program below the cursor position, **Backward** searches the program above the cursor position.

After you specify at least the code that you want to search for and the direction you want to search, press the function key for the search mode that you want to use:

Find Code ([F1])

Press **[F1]** to find the search term.

The control searches the program in the specified direction and highlights the first occurrence of the search term. Each time you press **[F1]**, the control searches for the next occurrence of the search term, in the search direction that you specified, until it reaches the end of the program.

Replace Code (**[F2]**)

After the search function finds an occurrence of your search term, you can press **[F2]** to replace that code with the contents of the **Replace** field.



NOTE:

*If you press **[F2]** with no text in the **Replace** field, the control deletes that occurrence of your search term.*

Find and Replace(**[F3]**)

Press **[F3]** instead of **[F1]** to start the find and replace operation. For each occurrence of your search term, press **[F3]** if you want to replace it with the text in the **Replace** field.

Replace All(**[F4]**)

Press **[F4]** to replace all of the search term occurrences in (1) step. You cannot undo this.

MODIFY Menu

The Modify menu has commands that let you make quick changes to an entire program, or to selected lines within a program.



NOTE:

*You cannot use **[UNDO]** to reverse Modify operations. The operations also automatically save the program. If you are not sure you want to keep the changes that you make, be sure to save a copy of the original program.*

- **Remove All Line Numbers:** Automatically removes all N-code line numbers from the program or the selected program blocks.
- **Renumber All Lines:** Automatically adds N-code line numbers to the program or the selected program blocks. Enter the line number you want to start with and the increment to use between line numbers, then press **[ENTER]** to continue or press **[UNDO]** to cancel and to return to the editor.
- **Reverse + And - Signs:** Changes positive values for selected address codes to negative, or negative values to positive. Press the letter key for the address codes that you want to reverse to toggle selections in the pop-up menu. Press **[ENTER]** to execute the command or **[CANCEL]** to return to the editor.